

Candidate Homework: Next Best Action Engine

Build a small, self-contained “Next Best Action” module for a generic CRM workflow.

The goal is to evaluate how you design an agent-like decision system that recommends what a user should do next for a customer or prospect, based on available signals and clear reasoning.

This exercise is intentionally project-agnostic. You should not assume any specific CRM product, sales process, or internal schema. Part of the assignment is doing your own research on CRM workflows and making reasonable product decisions.

What You Should Build

Create a module that takes in a customer/prospect record plus recent activity and returns the top recommended next actions.

Each recommendation should include:

- `action_type`
- `title`
- `reasoning`
- `confidence`
- `evidence`
- `risks_or_caveats`

Examples of actions:

- send follow-up email
- create call task
- schedule meeting
- pause outreach
- request more data
- escalate to account executive
- mark as low priority

You may define your own action taxonomy if it is coherent.

Required Inputs

Your system should work from a compact structured input such as:

- person/account record
- lifecycle or pipeline stage

- recent activities/interactions
- last-touch timing
- important missing fields
- consent / communication status
- any other CRM signals you believe are relevant

You may design the schema yourself.

Required Behavior

Your solution should:

- return the top 3 recommendations
- use structured output, not just free-form text
- explain why each recommendation was made
- include evidence used for each recommendation
- include a confidence score
- handle uncertainty explicitly
- avoid recommending actions that should obviously be blocked by missing consent, missing information, or contradictory signals

You do not need to build a full LLM workflow. Deterministic logic, scoring systems, lightweight planners, or hybrid approaches are all acceptable if well justified.

Important: Research Expectation

You are expected to do your own research on CRM workflows, sales follow-up patterns, and common next-step logic.

We are deliberately not giving you a detailed CRM playbook. We want to see:

- how you frame the problem
- which assumptions you make
- how you justify those assumptions
- whether you can translate messy business context into a clean decision module

Please document your product assumptions clearly.

AI Tooling

You are encouraged to use AI coding tools such as Claude Code, Codex, Cursor, Windsurf, or similar.

We want to see good judgment, not raw manual coding only.

Include a short `AI_USAGE.md` describing:

- which AI tools you used
- what you used them for
- where they were helpful
- where they were wrong or incomplete
- what you changed or rejected

Deliverables

Please submit:

- working code
- a short `README.md`
- tests
- `AI_USAGE.md`

Your `README.md` should briefly cover:

- problem framing
- assumptions about CRM behavior
- architecture and approach
- ranking / recommendation logic
- tradeoffs
- what you would improve next with more time

Suggested Scope

Keep it relatively self-contained. A good submission might include:

- a core recommendation engine
- a small API endpoint or CLI entry point
- sample input/output fixtures
- unit tests for ranking and guardrails

A simple UI is optional. If you add one, keep it lightweight.

Timebox

Target 2-3 hours.

We are not looking for a fully polished product. We are looking for clear thinking, sensible abstractions, and disciplined execution.

Evaluation Criteria

We will evaluate:

- quality of problem framing
- product judgment
- clarity of assumptions
- code quality
- correctness and test coverage
- structure of outputs
- handling of uncertainty and guardrails
- ability to use AI coding tools effectively without over-trusting them